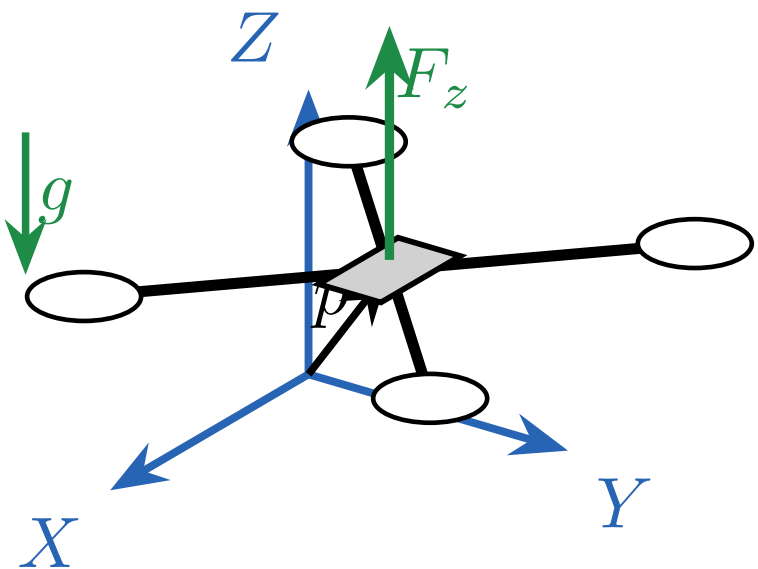

$$x = \begin{bmatrix} p \\ \phi \\ v \\ \omega \end{bmatrix} \in \mathbb{R}^{12}$$

$$p = (x, y, z), \quad \phi = (\phi, \theta, \psi)$$

$$u = \begin{bmatrix} F_z \\ M_x \\ M_y \\ M_z \end{bmatrix}$$

$$\dot{\phi} = T(\phi)\,\omega$$